

Red Hat OpenShift 4.16 has new features

Red Hat, Inc., the global leader in open source solutions, has announced new features and enhancements for Red Hat OpenShift, the hybrid cloud application platform powered by Kubernetes. Additionally, Red Hat Advanced Cluster Security Cloud Service is now generally available. These updates, part of the general release of Red Hat OpenShift 4.16, aim to simplify the development, connection, and security of various workloads, ensuring a consistent experience across applications and environments.

“Being able to more quickly modernise both infrastructure and applications are crucial capabilities to better support customer needs -- but this can’t be done overnight. As organisations make technology decisions in support of their AI strategies, both for today and tomorrow, they need platforms that set them up for future success while also balancing the demands of their legacy applications and environments. With Red Hat OpenShift 4.16, organisations can focus on both incremental improvements and modernisation. Red Hat OpenShift helps customers prepare their infrastructure for the demands of AI while managing and maintaining traditional, mission-critical applications and infrastructure like virtualised environments, all from a single platform,” said Mike Barrett, vice president and general manager, hybrid cloud platforms, Red Hat.

The latest enhancements to Red Hat OpenShift are designed to help organisations connect their diverse workloads and create a consistent management and deployment experience across the hybrid cloud. These enhancements ensure seamless integration and management, whether the application is a traditional workhorse or built to harness AI innovation.

enterprises, enabling more efficient insight extraction and data-driven decision-making at no additional cost. “HeatWave’s stunning pace of innovation continues with the addition of HeatWave GenAI,” said Edward Screven, Oracle’s chief corporate architect. “Integrated and automated AI enhancements now allow developers to build rich generative AI applications faster, without requiring AI expertise or moving data.”



Vijay Sundhar, CEO of SmarterD, highlighted the benefits of HeatWave GenAI, stating, “Support for in-database LLMs and vector creation leads to a significant reduction in application complexity and predictable inference latency, all at no additional cost. This democratises generative AI and boosts productivity.

Tensor unit efficiency data revealed for Semidynamics’ AI IP

Semidynamics, a European RISC-V custom core AI specialist, has revealed tensor unit efficiency data for its ‘all-in-one’ AI IP running a LLaMA-2 7B-parameter large language model (LLM). This new AI architecture integrates these elements into a single processing unit, combining a RISC-V core, a tensor unit for matrix multiplication, and a vector unit for activation-like computations. This architecture is DMA-free, uses a single software stack based on ONNX and RISC-V, and offers direct connectivity between elements. The result is higher performance, lower power consumption, better area utilisation, and easier programming, reducing overall development costs. The tensor and vector units, controlled by a flexible CPU, can deploy any current or future AI algorithm, protecting customer investments.

Large language models (LLMs), key in AI applications, are dominated by self-attention layers involving matrix multiplications, matrix transpose, and SoftMax activation functions. In Semidynamics’ all-in-one solution, the tensor unit handles matrix multiplication, while the vector unit manages transpose and SoftMax. Shared vector registers between these units eliminate expensive memory copies, ensuring zero latency and zero energy expenditure in data transfers. Efficient data fetching from memory into vector registers is achieved through Semidynamics’ Gazzillion Misses technology, which supports numerous in-flight cache misses.

The company tested the full LLaMA-2 7B-parameter model using its ONNX Run Time Execution Provider, calculating tensor unit utilisation for MatMul layers. The stats indicate utilisation above 80% for most shapes, even under challenging conditions. They also demonstrate tensor unit efficiency for large matrix sizes,